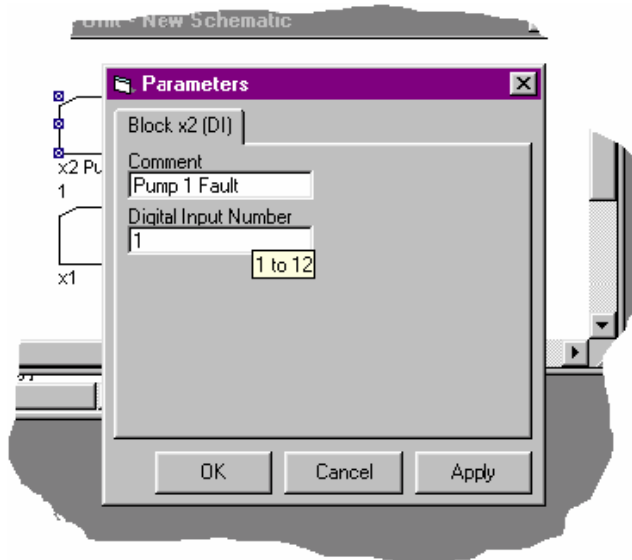


1.3.5. Entering Parameters

Some function blocks require parameters to be entered to work. The sloped top-left corner can easily identify these blocks. The digital input and output blocks on the previous page are examples of this.

- ☐ Double click one of the digital input blocks on the left hand side of the schematic.



The parameters window should now open and show the two parameters to be entered.

If you are unsure about a parameter, look in the documentation (Bulletin NXAB-1001) for a complete description of the function.

Another useful feature is the 'tip text'. If you move the mouse pointer over the edit field but do not click it, a tip will appear. This will give the allowable range for the parameter if applicable.

The first parameter here is called 'Comment'. This is text used to describe a function purely for documentation purposes.

- ☐ If you wish, you may leave comment blank, but for illustration enter some text such as 'Pump 1 Fault' in this example.

The second parameter here is the actual digital input number to use. This is how the system knows which terminals to read the input voltage from.

- ☐ Enter a number '1' for the input number parameter.

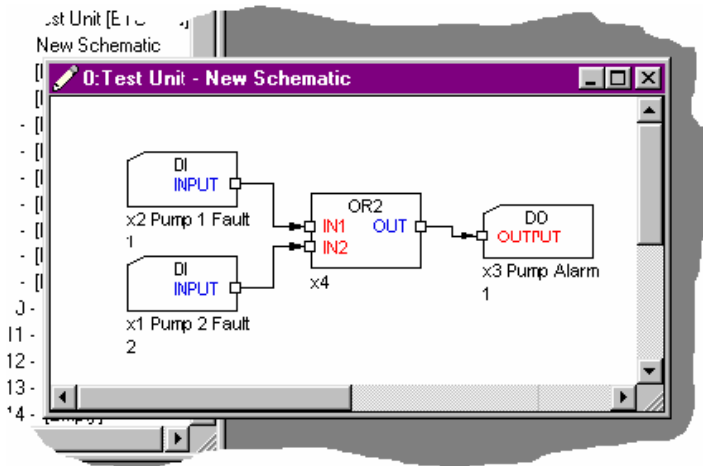
Pressing 'OK' will transfer these new parameters to the block and close the parameters window. Pressing 'Cancel' will discard the parameters you've just entered. Pressing 'Apply' will transfer the new parameters to the block and leave the parameters window open. This is useful when entering / reviewing the parameters for several blocks.

- ☐ Click 'Apply'.
- ☐ While leaving the parameters window open, click on the other digital input block. You may need to move the parameters window if it is covering this block. In this example, the block is partially covered.

The parameters window automatically updates itself with parameters from the new block. In this case, they are all blank.

- ☐ Enter something for the comment field - 'Pump 2 Failed', for example, and a value of 2 for the digital input number. Click 'Apply' once more.
- ☐ Leaving the parameters window open, click on the digital output block, and enter a comment field 'Pump Alarm' for example, and a digital output number of 1.
- ☐ Click OK to apply the parameters and close the window.

The schematic should now look similar to this.



Note that the parameters are listed by the function block in the order that they appear in the parameters window. The look can be a bit unusual but is vital for documentation and debugging purposes.

1.3.6. Checking your schematic- text compiling

With your schematic page active, choose 'Project' - 'Test compile schematic' from the menu. This will perform a 'test compile' of your page. The result will tell you if everything's okay, or if you've not entered a parameter or if there is some other problem with your schematic.

Hopefully, all is ok. If not check the block concerned. The error message will tell you the block x-number of any erroneous blocks.

Checking your schematic - IO Usage

While your schematic may compile ok, you may wish to check a few more things. For example you may have used the same output twice, or the same input for multiple unrelated items.

For additional information see [The IO usage window](#) (when actually working in Abacus).

1.3.7. Linking and uploading your program

In order to upload the program to the Fireye PPC6000 or NX6100 control, Fireye's Comfire 2 version 2.0 or above must be running on your PC. For a complete description of Comfire 2 capabilities refer to Fireye bulletin NEX-1502.

- ☐ Before uploading, it may be a good idea to save the project file. Click  on the top toolbar, or choose 'File' - 'Save project' from the menu.

- If Fireeye Comfire 2 is not already running on your system, start it up. You can do this from the



menu. You'll find it under programs - Comfire 2.

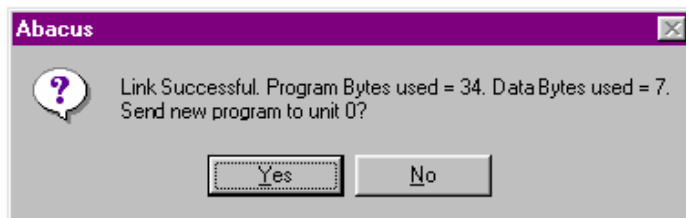
After Comfire 2 starts, it should see the units connected to it - if it has been connected. Click the 'Scan Units' button. The unit address should be the same as that used for your program.

- Switching back to abacus, ensure that the schematic page is open and selected (click within the schematic), and choose 'Project' - 'Make program (Link schematics)' from the menu.

Abacus will now open all the schematic pages in your project that contain programs for the same unit as the current schematic page (one in this case).

When the compile / link process has finished - almost immediately in this case, this message will appear telling you how much memory (see [Program size limits](#)) has been used. Click 'Yes' to send the program to the control, or 'No' if this was just a 'dry run'.

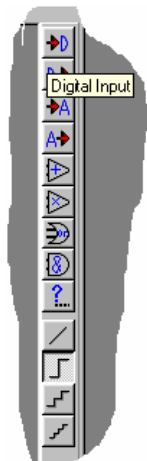
NOTE: THE CONTROL MUST BE IN FULL COMMISSION MODE TO ACCEPT PROGRAM UPLOAD.



If the download completes with no errors, your program is now installed the control as the 'user program'. There can only be one 'user program' installed in a control. When you upload your program, you will overwrite any existing user program. See [Choosing the Running Program](#) to make your new program the active program (i.e. to run it).

1.4. The schematic toolbar

The function block toolbar can be used whenever a schematic page is open and active. If you are unsure what a button does, hold the mouse pointer over it and wait for the tip to come up as shown here.



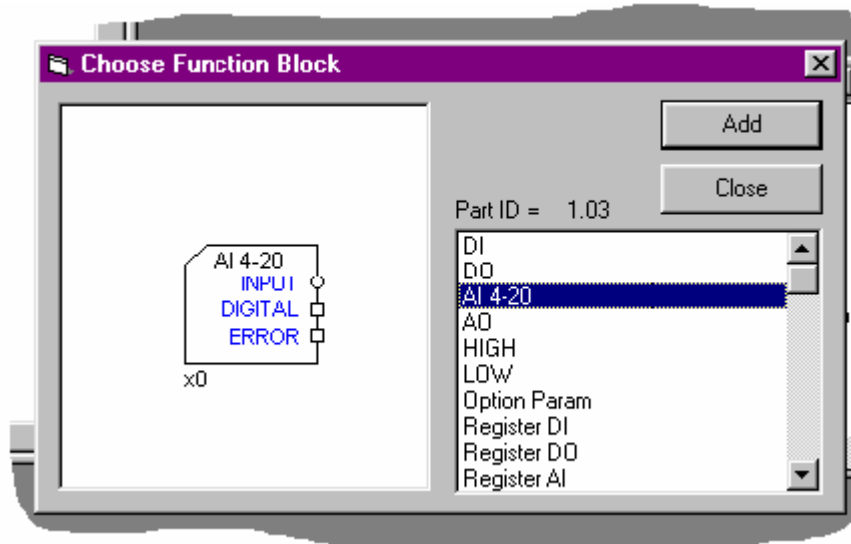
The top section of buttons will place selected function blocks on your schematic. The actual buttons shown may vary from those shown here, but the idea is the same.

This button  is the 'other block' button. This will open up a dialog window allowing you to select any function block from the database. See [Place Function blocks](#).

These 'line' buttons are used when editing the positions of the lines that link the blocks on the screen. The next link to be drawn will be in the style selected here. Also, you can use these to change the style of a currently selected link.

1.5. Place Function blocks

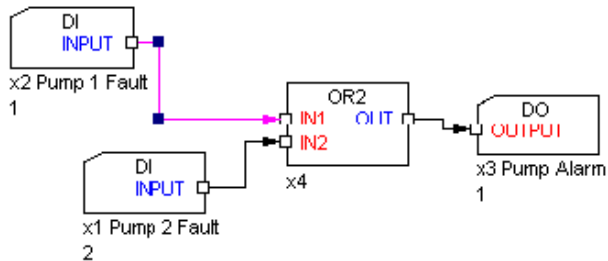
When the  button is clicked on the schematic toolbar, this window will open.



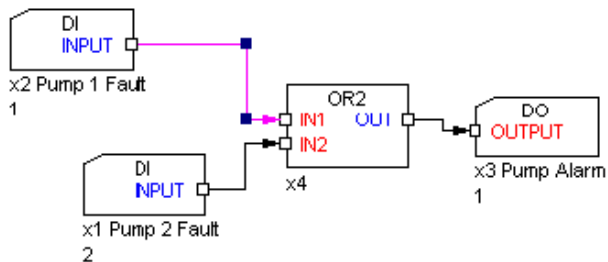
By clicking in the list you can preview all available function blocks. Click 'Add' to add the currently shown block to your schematic. To get help on the currently selected block, press F1.

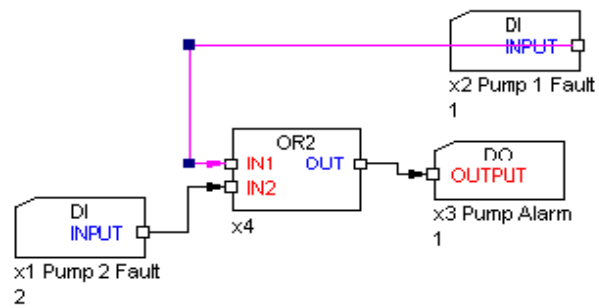
1.6. Moving and editing the link lines

You can click on a link to select it. It will turn pink, and handles will appear on the corners of that link like this.



You can drag these handles to move the middle section of the link.



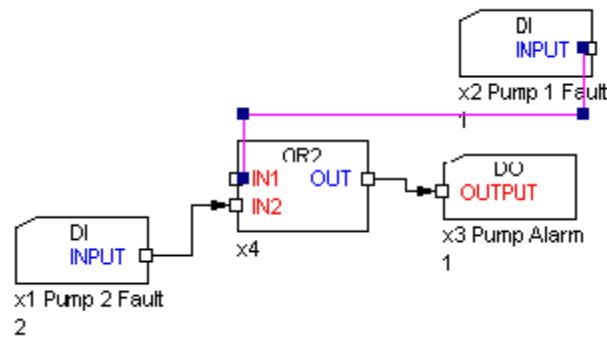


In this case, block x2 has been moved to the right, and now the link goes backwards through the block itself. There is nothing wrong, the program is the same, however, the appearance is unclear.

If you grab and drag the line handles in this case, you will not be able to improve the appearance.

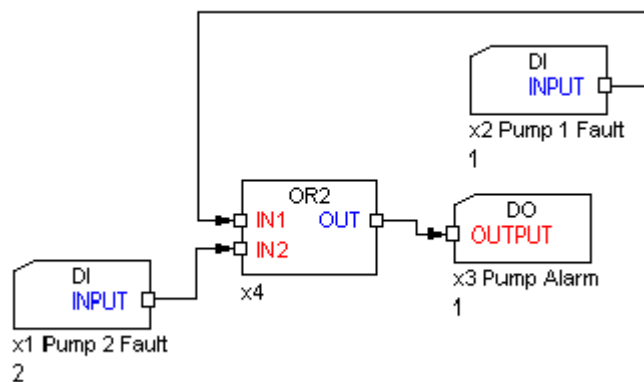
It is here that you need to use the line style buttons on the schematic toolbar.

- With the link selected (i.e. pink), click the  button on the schematic tool bar. The schematic will change to the following:



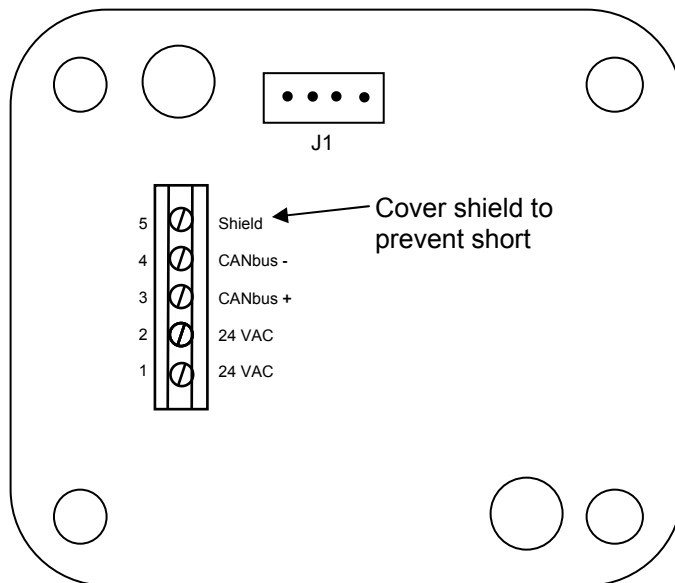
While not much clearer, you now have more handles to drag and can make the schematic look sensible once more.

The finished article may look like this:



11 Circuit board connections

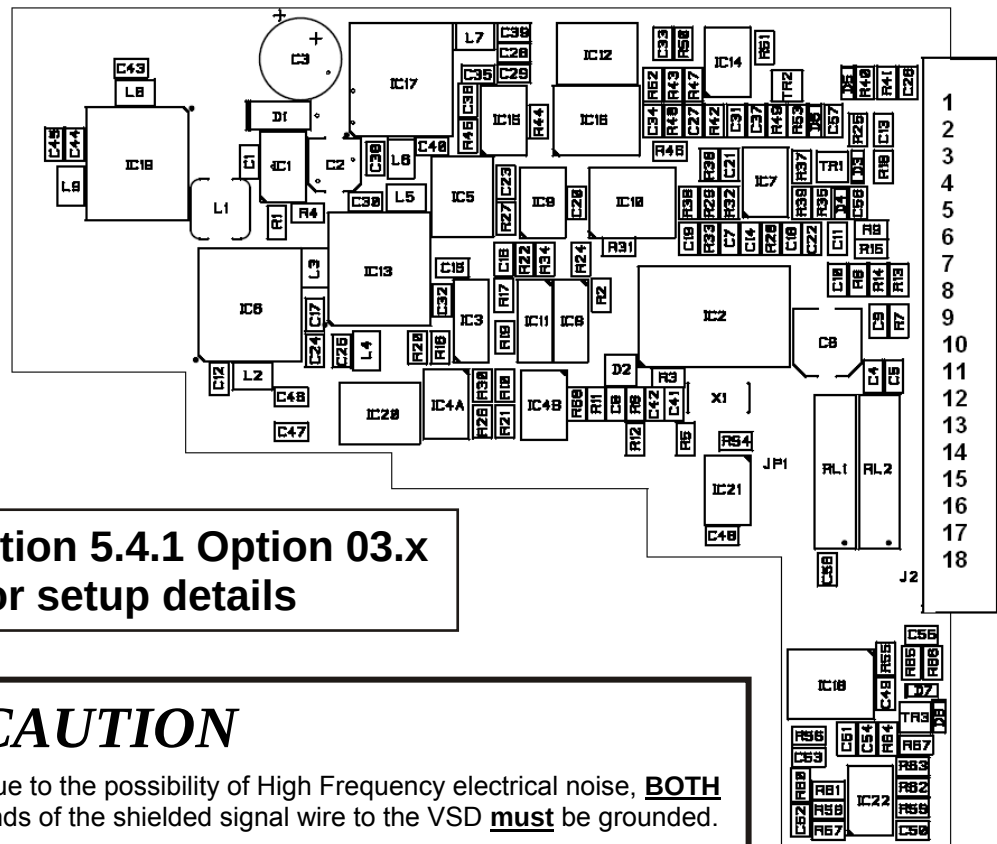
11.1 Inlet Air Temperature Sensor (Optional)



NOTE: Insulate Shield (Terminal 5) to prevent accidental shorting to pins on circuit board (J1)

NXIATS
Inlet Air Temperature Sensor

11.2 PPC6000 Variable Speed Drive Daughter Board P/N NXDBVSD



CAUTION

Due to the possibility of High Frequency electrical noise, **BOTH** ends of the shielded signal wire to the VSD **must** be grounded.

- 4-20mA Output Channel 1 – PZ 1
- 4-20mA Output Channel 1 + PZ 2
- 4-20mA Output Channel 2 – PZ 3
- 4-20mA Output Channel 2 + PZ 4
- 4-20mA Output Channel 3 – PZ 5
- 4-20mA Output Channel 3 + PZ 6
- Counter Input 1 PZ 7
- +12Vdc PZ 8
- Counter Input 2 PZ 9
- +12Vdc PZ 10
- Counter Input 3 PZ 11
- Input Channel 1 – PZ 12
- Input Channel 1 ± PZ 13
- Input Channel 2 + PZ 14
- Relay 7 common PZ 15
- Relay 7 normally open PZ 16
- Relay 8 common PZ 17
- Relay 8 normally open PZ 18

The cabling for all of these terminals must be 2-conductor (core) overall screened PVC insulated 24 AWG (7/0.2mm). Since this cable may be run in conduit with high voltage wiring, its voltage rating must exceed the maximum voltage carried by any other cable connected to the control or run in the same conduit.

The counter inputs are for use with encoder(s) for speed feedback from the devices being controller by the inverter(s) connected to the 4-20mA output(s) on this daughter board.

The analog outputs are isolated but the analog inputs are not and therefore special precautions must be taken when using these inputs.

The inputs are for 4-20mA signals **must** be isolated as the second input connects 'on-top' of the first to ensure 'cross-talks' will be detected. Terminal PX13 alternates ± to be shared with PZ12 and PZ14. This "switching" detects a direct link between input 1 and input 2 which is not allowed.

The function of relay 1 & 2 can be set by option parameter, but please note the maximum rating of these relays is 200mA at 50V.

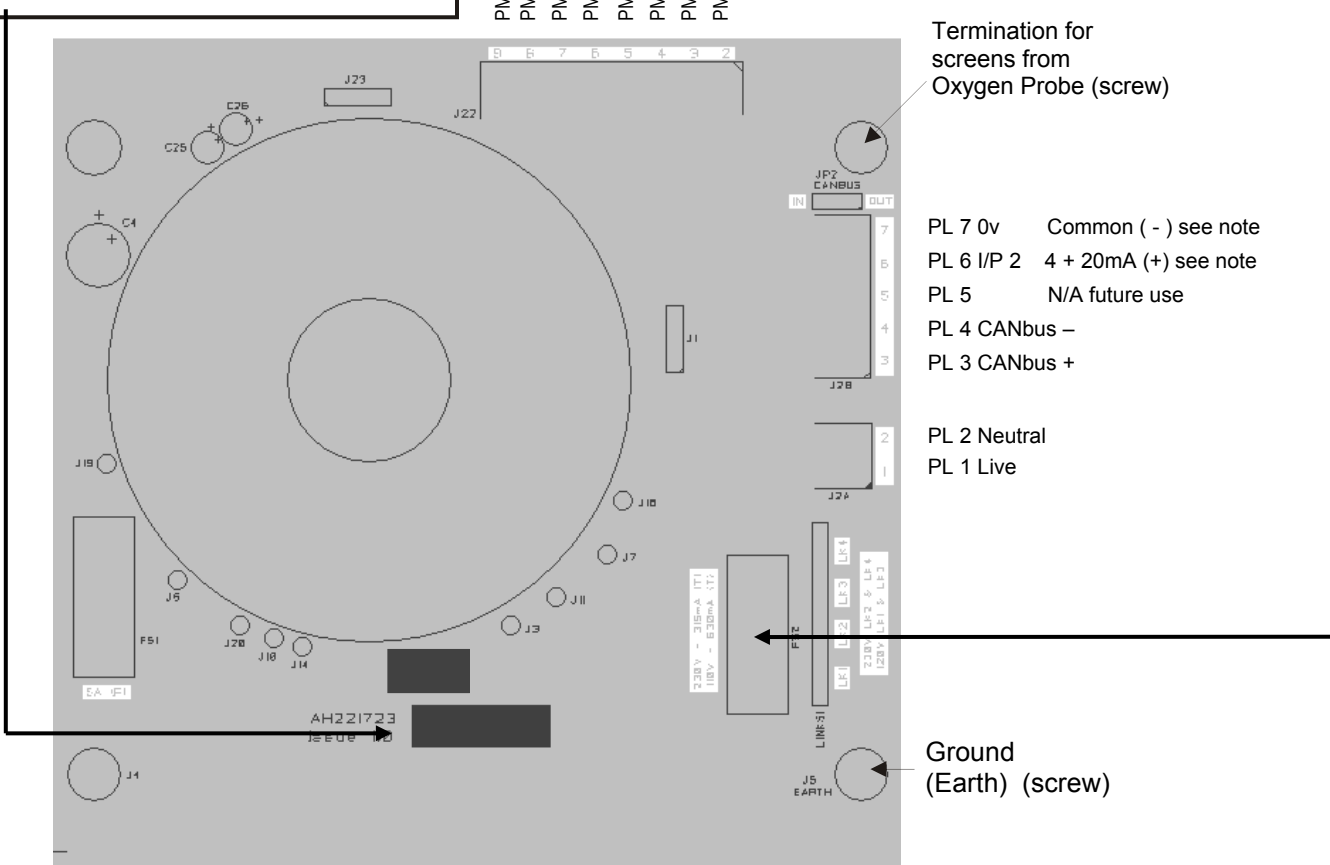
Incorrect connection may damage or destroy the units being connected.

11.3 Oxygen probe connection (optional Oxygen Probe Interface unit)

Bussman S504 series
PN GMA-5
5 AMP (Fast Acting)
(to protects transformer primary DC circuit)
Alternate Source:
Mouser Electronics : PN 504-GMA-5

PM 9- PROBE HEATER >24VAC
PM 8- PROBE HEATER >24VAC
PM 7- FLUE TEMP. THERMOCOUPLE +
PM 6-FLUE TEMP. THERMOCOUPLE -
PM 5 - CELL TEMP. +
PM 4 - CELL TEMP. -
PM 3 - CELL OUTPUT +
PM 2 - CELL OUTPUT -

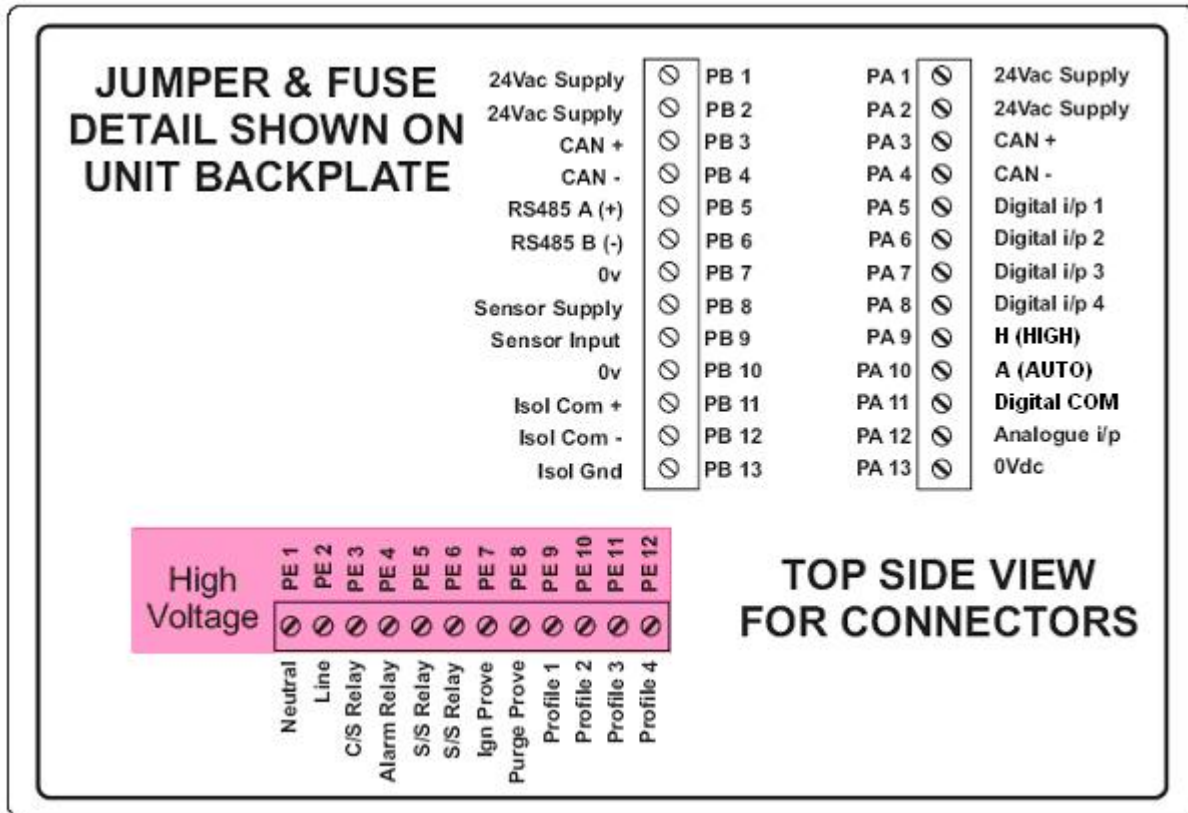
Bussmann - S504 series
P/N GMD-630 (TIME Delay) for 115V
P/N GMD-315 (TIME Delay) for 230V
(to protect 24V circuit)
Alternate Source::
Mouser Electronics



NOTE: 4-20mA O₂ input from non-Fireye O₂ probe. See sections 30.1 and 30.2

NX02INT

11.4 PPC6000



Note: All wiring to terminals PA & PB are low voltage and must be braided shielded wire per table 2.6.1A. Wiring to terminals “PE” is line voltage. The maximum wire size is 16AWG (19/.3mm).

Note: PE7 & PE8 provide purge and low fire position signals to the flame safeguard control. These outputs **MUST NOT** have a load greater than 30mA (i.e. relays, lamp, etc.), damage to the PPC6000 will result.



This manual may be downloaded from the Fireeye web site at: www.Fireeye.com
Click on "Bulletins" or search by bulletin number.

NOTICE

When Fireeye products are combined with equipment manufactured by other and/or integrated into systems designed or manufactured by others, the Fireeye warranty, as stated in its General Terms and Conditions of Sale, pertains only to the Fireeye products and not to any other equipment or to the combined system or its overall performance.

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FIREYE
3 Manchester Road
Derry, New Hampshire 03038 USA
www.fireeye.com

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